Spatial–Spectral Cross Mamba Network for Hyperspectral and Multispectral Image Fusion

Huapeng Wu, Zhiyu Sun, Jiaqiang Qi, Tianming Zhan, *Member, IEEE*, Yang Xu, *Member, IEEE*, and Zhihui Wei, *Member, IEEE*

***Abstract*—Currently, hyperspectral and multispectral image fusion methods based on local and global feature learnings (e.g., convolutional neural network (CNN) and transformer) have achieved promising results. However, as the core part of the transformer, the computational cost of the self-attention is quadratic with the image size, which severely limits its practical application. In this article, we propose a spatial–spectral cross mamba (SSCM) network for hyperspectral and multispectral image fusion. By using the mamba structure, our model can obtain long-range spatial–spectral information with less computational complexity in comparison with the transformer structure. Specifically, we introduce an SSCM block to facilitate the interaction between hyperspectral and multispectral features, effectively enhancing the spatial–spectral feature representation ability of the network. In addition, a cross-scale spatial–spectral learning module (CSSSM) based on the U-shaped structure is proposed to effectively extract the long-range high-frequency feature information at different scales. The extensive experimental results demonstrate that our method achieves comparable performance in comparison with some state-of-the-art image fusion methods.**

***Index Terms*—Hyperspectral image (HSI), image fusion, Laplacian, Mamba.**

I. Introduction

HYPERSPECTRAL images (HSIs) typically consist of numerous sequential, narrowband spectral bands that span from visible light to the near-infrared spectrum [1], [2], [3], which is widely used in the fields of image classification [4], [5], [6], target detection [7], [8], [9], change detection [10], [11], [12], and anomaly detection [13], [14], [15]. However, because of the constraints of imaging systems and equipment [16], [17], [18], the obtained remote sensing images are often of low spatial resolution or spectral resolution, which hinders the widespread application of HSI. HSI super-resolution is one of the important image enhancement methods [19], [20], [21], [22], which can reconstruct high-resolution HSIs. HSI super-resolution mainly consists of single-input super-resolution and hyperspectral and multispectral image fusion.

In recent years, the rapid development of deep learning techniques and the hyperspectral and multispectral image fusion methods based on deep convolutional neural networks (CNNs) have achieved significant success. Yang et al. [23] proposed two-branch CNNs for low-resolution HSI (LR-HSI) and high-resolution multispectral image (HR-MSI) fusion. The proposed network can extract spectral information from an HSI and spatial information from a multispectral image, respectively, achieving good image fusion results. Hu et al. [24] presented a deep neural network architecture that leverages CNNs to extract high-frequency components from HR-MSI and LR-HSI, thereby effectively enhancing the spatial structure details of the reconstructed high-resolution HSI. Palsson et al. [25] introduced a 3D-CNN for LR-HSI and HR-MSI fusion. The proposed network effectively reduces the computational complexity of the network by reducing the dimension of the HSI before image fusion. Liu et al. [26] proposed a multiscale gradient spatial attention module to extract the spatial edge details of HSIs at multiple scales. The proposed network adopts a mutual learning strategy to allow knowledge transfer across different input conditions and improve network optimization.

To overcome the local limitations of CNN, the transformer was introduced to learn long-range context information, which has been successfully applied in various fields. Bose et al. [27] proposed a transformer-based HSI and LiDAR data fusion network, which effectively improved the interaction of multimodal data. Hu et al. [28] introduced a transformer-based HSI and MSI fusion structure (Fusformer), which achieved good global feature extraction results. To capture spectral image features at different levels, Deng et al. [29] introduced a pyramid shuffle-and-reshuffle transformer network (PSRT) for LR-HSI and HR-MSI fusion. The introduced network utilizes a pyramid shuffle-and-reshuffle structure with windowed self-attention to extract image features. To fully integrate the LR-HSI and HR-MSI images, a two-branch cross transformer architecture [30] was proposed for image fusion. This network can effectively enhance the complementary features between HSI and MSI. Li et al. [31] proposed a cross-scale nonlocal attention network for HSI super-resolution, which effectively extracts the nonlocal spatial–spectral features of the image.

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Huapeng Wu, Zhiyu Sun, and Jiaqiang Qi are with the School of Computer Science, Nanjing Audit University, Nanjing 211815, China.

Tianming Zhan is with Jiangsu Key Construction Laboratory of Audit Information Engineering and the School of Computer Science, Nanjing Audit University, Nanjing 211815, China (e-mail: ztm@nau.edu.cn).

Yang Xu and Zhihui Wei are with the School of Computer Science and Engineering, Nanjing University of Science and Technology, Nanjing 210094, China.

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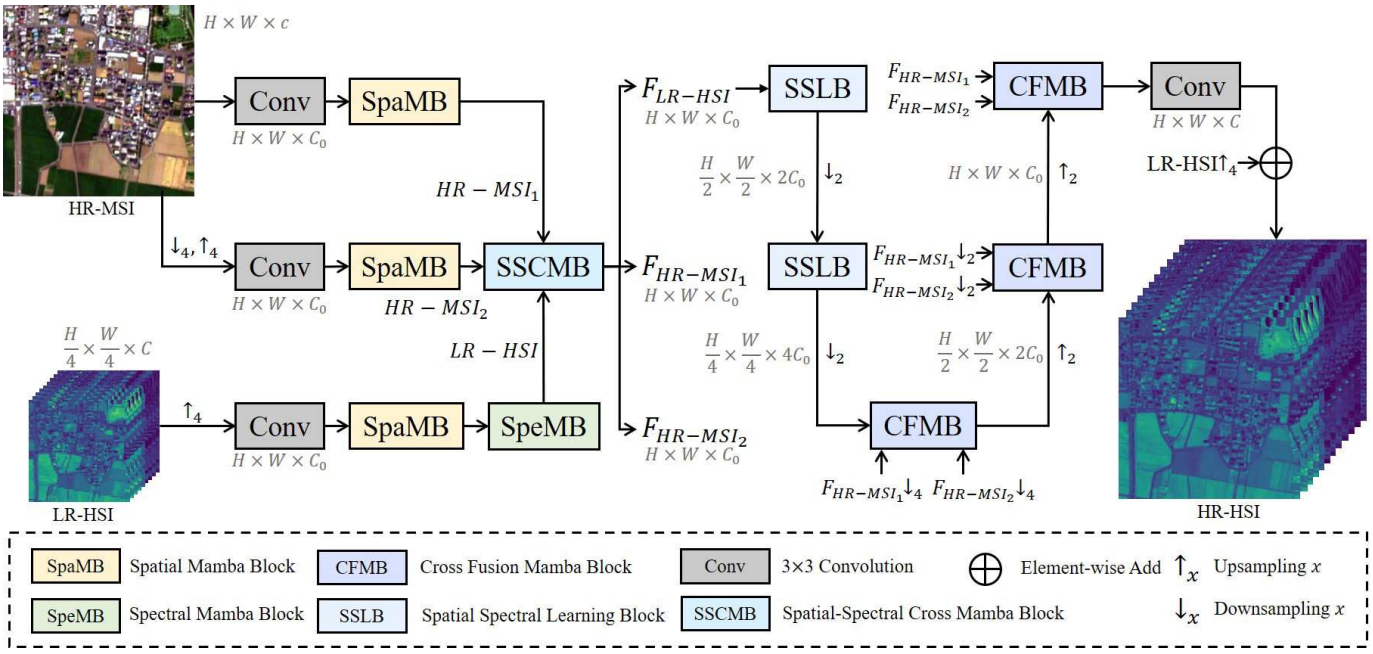


Fig. 1. Architecture of the proposed SSCM network.

Sun et al. [32] proposed a multiscale cross-fusion transformer network to reconstruct a high-resolution HSI. The network employs a multihead cross-attention module and a progressive reconstruction module to extract spatial–spectral features at various dimensions, which significantly improves image fusion performance. Although the transformer-based image fusion method achieves good performance, its high computational complexity limits its application capabilities.

Recently, deep networks based on mamba can obtain long-range context information using linear complexity, which is widely used in the field of computer vision. Specifically, Zhu et al. [33] first applied the mamba to computer vision (vision mamba) and achieved better performance than some transformer-based networks in image classification, object detection, and semantic segmentation tasks. Considering the 1-D scanning property of mamba, Liu et al. [34] proposed a VMamba network that can preserve the 2-D features of the image. The introduced visual state space (VSS) module can cross-scan from different spatial dimensions to extract global contextual information from different sources and angles, achieving good experimental performance in various vision visual tasks. He et al. [35] proposed an effective pan-sharpening network (Pan-Mamba) by using a state-space model (SSM), showing superior fusion results. To thoroughly investigate the capabilities of the mamba, Peng et al. [36] proposed Fusion-mamba, a mamba module based on U-Net to effectively extract and fuse spatial–spectral features, achieving impressive results in image fusion tasks. Zhao et al. [37] introduced a spatial–spectral interactive CNN-Mamba network to interactively reconstruct an image by using the spatial and spectral super-resolution module, which effectively prevents spectral distortion and enhances the spatial and spectral details of the fused image. Zhang et al. [38] introduced a cross-frequency enhanced spatial–spectral fusion network for hyperspectral and multispectral image fusion. The proposed network can effectively explore the frequency and spatial–spectral feature information of the image by using the fast Fourier transform (FFT) and SSM.

Although the above mamba-based networks achieve good HSI reconstruction results, they do not effectively extract and fuse global spatial and spectral features of hyperspectral and multispectral images, which leaves room for further improving image fusion results. In this article, we propose a spatial–spectral cross mamba (SSCM) network for hyperspectral and multispectral image fusion. As illustrated in Figs. 1 and 2, the input HR-MSI and LR-HSI are first input to an SSCM block (SSCMB) (as illustrated in Fig. 3) to learn the long-range spatial and spectral dependences between them to improve the spectral representation of the HR-MSI and the spatial representation of the LR-HSI, respectively. Subsequently, the enhanced feature is passed into a U-shaped network, which consists of a spatial–spectral learning block (SSLB) and a cross-fusion mamba block (CFMB) used to extract important long-range spatial–spectral edge and texture information at different scales. Specifically, in the encoding stage, SSLB (as illustrated in Fig. 4) contains Laplacian operator, spatial mamba, and spectral mamba, which can effectively learn the informative global edge and texture detail features in the spatial and spectral dimensions. In the decoding stage, CFMB is utilized to extract long-range feature information and improve the spatial representation ability of the model with the help of the HR-MSIs. The main contributions of our SSCM are as follows.

1) We propose a novel SSCM network for hyperspectral and multispectral image fusion. The proposed network can effectively extract long-range spatial–spectral features at different scales.

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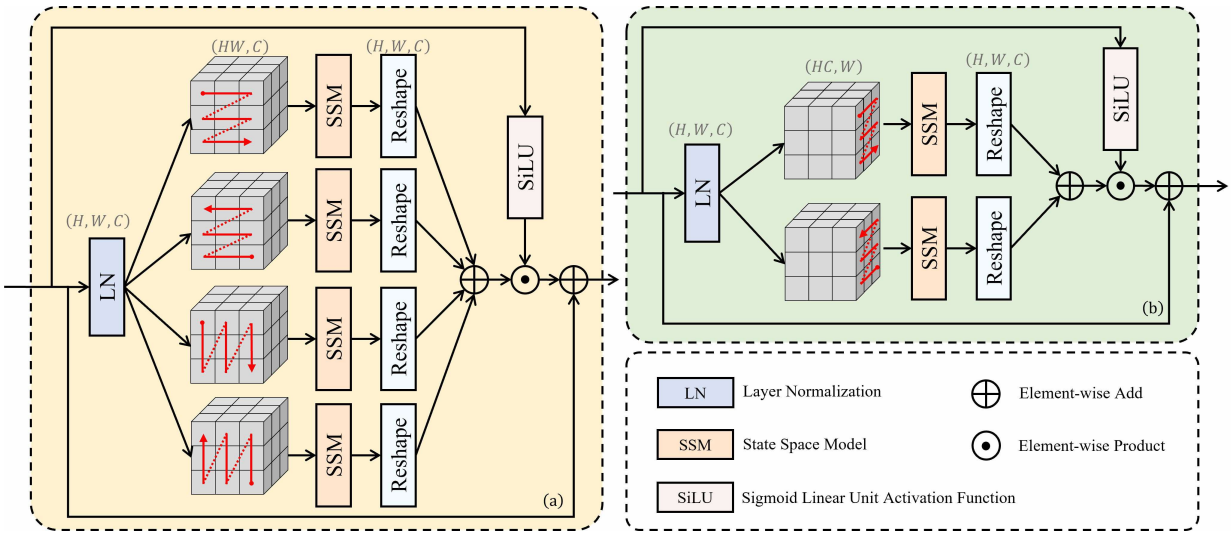


Fig. 2. (a) Structure of SpaMB. (b) Structure of SpeMB.

2) We propose an SSCMB to integrate the spatial and spectral feature information of the HR-MSI and LR-HSI.
3) We introduce a U-shaped architecture with SSLB and CFMB, which can effectively learn the informative global contextual texture features at different scales. The experimental results show that the proposed network achieves comparable fusion performance in comparison with some state-of-the-art HSI and MSI fusion methods.

This article is organized as follows. Section II describes the details of our SSCM network. Section III presents the experimental results and ablation analysis. Finally, the conclusion is provided in Section IV.

II. Method

In this section, Section II-A describes the overall structure of the proposed SSCM, Section II-B explains the principles of mamba, Section II-C provides details about the SSCMB, and Section II-D presents the cross-scale spatial–spectral learning module (CSSSM).

A. Overall Framework

As shown in Fig. 1, the proposed method mainly consists of three components: shallow feature fusion, deep feature learning, and reconstruction module. Specifically, $X \in \mathbb{R}^{h\times w\times C}$ and $Y \in \mathbb{R}^{H\times W\times c}$ denote the input LR-HSI and HR-MSI, h and H represent the heights of the two images, w and W represent their widths, and c and C indicate the number of spectral channels. The final output HR-HSI is defined as $I_{\mathrm{SR}} \in \mathbb{R}^{H\times W\times C}$. To alleviate the misalignment of the upsampling LR-HSI and HR-MSI, we perform downsampling and upsampling operations on HR-MSI before feature fusion. First, 3×3 convolution filters are applied to extract the initial features from three input images. Then, the initial features are fed into the spatial mamba and spectral mamba [as shown in Fig. 2(a) and (b)] to extract long-range spatial and spectral features, respectively. An SSCMB is used to fuse the spatial and spectral features of the LR-HSI and HR-MSI and improve the spatial and spectral representations of the network. The above spatial–spectral feature extraction process can be expressed as

$$\begin{aligned}\text{LR-HSI} &= \text{SpeMB}\left(\text{SpaMB}\left(F_{\text{LR}}\right)\right)\\ \text{HR-MSI}_1 &= \text{SpaMB}\left(F_{\text{HR}_1}\right)\\ \text{HR-MSI}_2 &= \text{SpaMB}\left(F_{\text{HR}_2}\right)\end{aligned} \tag{1}$$

where SpaMB$(\cdot)$ represents the spatial mamba block, and SpeMB$(\cdot)$ is the spectral mamba block. Subsequently, LR$-$HSI, HR$-$MSI$_1$, and HR$-$MSI$_2$ are fed into the SSCMB to facilitate the interaction of spatial and spectral features. The module can be written as

$$\begin{aligned}&F_{\text{LR-HSI}}, F_{\text{HR-MSI}_1}, F_{\text{HR-MSI}_2}\\ &= \text{SSCMB}\left(\text{LR-HSI}, \text{HR-MSI}_1, \text{HR-}MSI_2\right)\end{aligned} \tag{2}$$

where SSCMB$(\cdot)$ represents the spatial–spectral cross mamba block, and $F_{\text{LR-HSI}}$, $F_{\text{HR-MSI}_1}$, and $F_{\text{HR-MSI}_2}$ are the enhanced LR-HSI and HR-MSIs. Finally, the enhanced features are fed into the U-shaped network to further learn important spatial–spectral feature information at different scales. The proposed network can be written as

$$I_{\text{SR}} = f_\theta\left(X, Y\right) + X\uparrow_4 \tag{3}$$

where θ represents the training parameters of the proposed network. $X\uparrow_4$ represents the bicubic linear interpolation upsampling operation with a factor of 4.

The $L1$ loss function is adopted to measure between the super-resolved (SR) images and the ground-truth (GT) images. The $L1$ loss function is defined as follows:

$$\mathcal{L}_1 = \frac{1}{K}\sum_{i=1}^{K}\left|SR_i - \text{GT}_i\right|_1 \tag{4}$$

where K denotes the quantity of training samples, and SR_i and GT_i denote the ith generated HR-HSI and the corresponding reference image.

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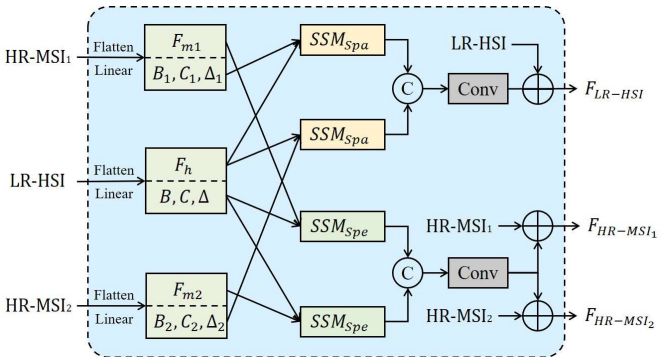


Fig. 3. Structure of SSCMB.

B. Mamba

The SSM is derived from modern control theory. Mamba [39] is a structured SSM that maps the 1-D input sequence $x(t)$ to the output sequence $y(t)$ through the cyclic update of the hidden state $h(t)$. This property can be written as a linear ordinary differential equation (ODE)

$$\begin{aligned} h'(t) &= Ah(t) + Bx(t) \\ y(t) &= Ch(t) \end{aligned} \tag{5}$$

where A represents the state matrix, and B and C are the mapping matrices. By incorporating a time-scale parameter Δ and zero-order hold (ZOH) technique, the discrete variables $\bar{A}$ and $\bar{B}$ are obtained by transforming the continuous variables A and B. These discrete variables are specified as follows:

$$\begin{aligned} \bar{A} &= e^{\Delta A} \\ \bar{B} &= (\Delta A)^{-1}\left(e^{\Delta A} - I\right) \cdot \Delta B. \end{aligned} \tag{6}$$

Thus, (5) can be rewritten in the subsequent discrete form

$$\begin{aligned} h_t &= \bar{A}h_{t-1} + \bar{B}x_t \\ y_t &= Ch_t. \end{aligned} \tag{7}$$

In addition, (7) can also be expressed using convolution

$$\begin{aligned} y &= \bar{K} * x \\ \bar{K} &= \left(C\bar{B}, C\bar{A}\bar{B}, \ldots, C\bar{A}^{L-1}\bar{B}\right) \end{aligned} \tag{8}$$

where $\bar{K}$ denotes the convolution kernel, L represents the sequence length of the input x, and * is the convolution operation.

C. Spatial–Spectral Cross Mamba Block

To effectively integrate the complementary feature information of the LR-HSI and HR-MSI, we propose an SSCMB to learn long-range spatial–spectral features. As illustrated in Fig. 3, the input LR-HSI, HR-MSI$_1$, and HR-MSI$_2$ denote the LR-HSI features, the HR-MSI features, and the continuous downsampling and upsampling of the HR-MSI, respectively. Considering that the feature information is mainly contained in the representation coefficients B, C, and Δ, we introduce an SSCMB to interact with different feature representation coefficients to achieve effective feature fusion. Specifically, the spatial feature coefficients of HR-MSI$_1$ and HR-MSI$_2$ and the sequence of the flattened spectral feature LR-HSI are fed into the spatial SSM [i.e., $\mathrm{SSM}_{\mathrm{Spa}}(\cdot)$] to improve the spatial representation ability of the LR-HSI feature LR-HSI. Similarly, the spectral feature coefficients of LR-HSI and the sequence of the flattened spatial features HR-MSI$_1$ and HR-MSI$_2$ are fed into the spectral SSM [i.e., $\mathrm{SSM}_{\mathrm{Spe}}(\cdot)$] to supplement the spectral information of the HR-MSI features HR-MSI$_1$ and HR-MSI$_2$. The SSCMB can be expressed as

$$\begin{aligned} f_1 &= \mathrm{SSM}_{\mathrm{Spa}}(\text{LR-HSI}, \text{HR-MSI}_1) \\ f_2 &= \mathrm{SSM}_{\mathrm{Spa}}(\text{LR-HSI}, \text{HR-MSI}_2) \\ f_3 &= \mathrm{SSM}_{\mathrm{Spe}}(\text{HR-MSI}_1, \text{LR-HSI}) \\ f_4 &= \mathrm{SSM}_{\mathrm{Spe}}(\text{HR-MSI}_2, \text{LR-HSI}) \\ F_{\text{LR-}HSI} &= \mathrm{Conv}_{1\times1}(\mathrm{Cat}(f_1, f_2)) + \text{LR-HSI} \\ F_{\text{HR-MSI}_1} &= \mathrm{Conv}_{1\times1}(\mathrm{Cat}(f_3, f_4)) + \text{HR-MSI}_1 \\ F_{\text{HR-}0_2} &= \mathrm{Conv}_{1\times1}(\mathrm{Cat}(f_3, f_4)) + \text{HR-MSI}_2 \end{aligned} \tag{9}$$

where $\mathrm{SSM}_{\mathrm{Spa}}(\cdot)$ represents the spatial mamba structure, $\mathrm{SSM}_{\mathrm{Spe}}(\cdot)$ represents the spectral mamba structure, $\mathrm{Cat}(\cdot)$ denotes the concatenation operation, and $\mathrm{Conv}_{1\times1}(\cdot)$ denotes the 1×1 convolution kernel. Specifically, take the $\mathrm{SSM}_{\mathrm{Spe}}(\mathrm{HR-MSI}_1, \mathrm{LR-HSI})$ as an example, and the calculation process of $\mathrm{SSM}_{\mathrm{Spe}}$ is as follows:

$$\begin{aligned} x_1 &= \mathrm{Flatten}_{\mathrm{Spe}}(\text{HR-MSI}_1) \\ x_2 &= \mathrm{Flatten}_{\mathrm{Spe}}(\text{LR-HSI}) \\ A_1 &= \mathrm{Parameter}_{A_1} \\ A_2 &= \mathrm{Parameter}_{A_2} \\ B_2 &= \mathrm{Proj}_B(x_2) \\ C_2 &= \mathrm{Proj}_C(x_2) \\ \Delta_2 &= s_\Delta\left(\mathrm{Parameter}_\Delta + \mathrm{Proj}_\Delta(x_2)\right) \\ \bar{A}_1 &= e^{\Delta_2 A_1} \\ \bar{B}_2 &= (\Delta_2 A_2)^{-1}\left(e^{\Delta_2 A_2} - I\right) \cdot \Delta_2 B_2 \approx \Delta_2 B_2 \\ h_t^1 &= \bar{A}_1 h_{t-1}^1 + \bar{B}_2 x_1 \\ y_t^1 &= C_2 h_t^1 \end{aligned} \tag{10}$$

where $\mathrm{Proj}_B(\cdot)$ and $\mathrm{Proj}_C(\cdot)$ denote the linear projection operations. s_Δ is a softplus activation function. The calculation process of $\mathrm{SSM}_{\mathrm{Spa}}$ is similar to $\mathrm{SSM}_{\mathrm{Spe}}$.

D. Cross-Scale Spatial–Spectral Learning Module

To extract important high-frequency information and long-range spatial–spectral features, we introduce a CSSSM with the SSLB and CFMB, which enables the network to effectively capture informative spatial and spectral features at different scales. As shown in Fig. 1, the proposed CSSSM is based on the U-shaped structure, which uses the SSLB and the CFMB as the feature encoder and decoder to extract the high-frequency and long-range features. Specifically, as illustrated in Fig. 4, our SSLB consists of three parallel branches. The middle branch aims to learn long-range spatial features by using the spatial mamba block (SpaMB). The purpose of the lower branch is to learn spectral information by using the spectral mamba block (SpeMB). The main purpose of the

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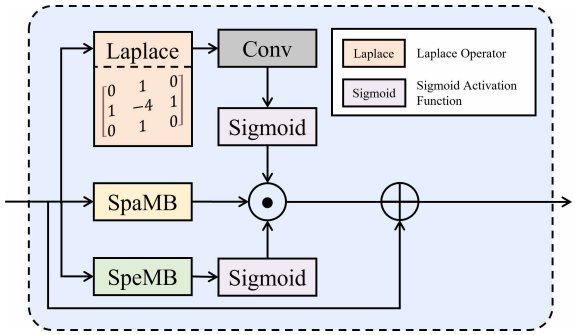


Fig. 4. Structure of SSLB.

upper branch is to improve the model's ability to learn edge structure features by leveraging the Laplacian operator. The proposed SSLB can be expressed as

$$\begin{aligned} F_{\text{Spa}}(x) &= \text{SpaMamba}(x) \\ F_{\text{Spe}}(x) &= \text{Sigmoid}\left(\text{SpeMamba}(x)\right) \\ F_{\text{Lap}}(x) &= \text{Conv}\left(\text{Sigmoid}\left(\text{Laplace}(x)\right)\right) \\ \text{SSLB}(x) &= F_{\text{spa}}(x) \odot F_{\text{spe}}(x) \odot F_{Lap}(x) + x \end{aligned} \tag{11}$$

where Sigmoid(·) represents the sigmoid activation function operations, Laplace(·) represents the Laplacian operator, and ⊙ represents the elementwise multiplication operation. During the upsampling process, we use the bicubic interpolation and a channel halving upsampling strategy. In addition, to improve the representation ability of the model, we utilize the CFMB to incorporate the HR−MSI$_1$ and HR−MSI$_2$ features from the previous stage to supplement spatial information. The relevant operations can be expressed as follows:

$$\begin{aligned} F_{\text{rc1}} &= \text{CFMB}\left(F_{\text{LR-HSI}}, F_{\text{HR-MSI}_1}\downarrow_4, F_{\text{HR-MSI}_2}\downarrow_4\right) \\ F_{\text{rc2}} &= \text{CFMB}\left(F_{\text{rc1}}\uparrow_2, F_{\text{HR-MSI}_1}\downarrow_2, F_{\text{HR-MSI}_2}\downarrow_2\right) \\ F_{\text{rc3}} &= \text{CFMB}\left(F_{\text{rc2}}\uparrow_2, F_{\text{HR-MSI}_1}, F_{\text{HR-MSI}_2}\right) \end{aligned} \tag{12}$$

where F_{rc3} represents the final result of the module, CFMB(·) represents the cross-fusion mamba block, $\uparrow_x$ denotes the bicubic linear interpolation upsampling operation with a factor of x, and $\downarrow_x$ represents the downsampling process with a factor of x. Take learning F_{rc1} as an example, and the CFMB can be expressed as

$$\begin{aligned} f_{1_1} &= \text{SSM}_{\text{Spa}}\left(F_{\text{LR-HSI}}, F_{\text{HR-MSI}_1}\downarrow_4\right) \\ f_{1_2} &= \text{SSM}_{\text{Spa}}\left(F_{\text{LR-HSI}}, F_{\text{HR-MSI}_2}\downarrow_4\right) \\ F_{\text{rc1}} &= \text{Conv}_{1\times1}\left(\text{Cat}\left(f_{1_1}, f_{1_2}\right)\right) + F_{\text{LR-HSI}} \end{aligned} \tag{13}$$

where $\text{SSM}_{\text{Spa}}(\cdot)$ is shown in 10, Cat(·) denotes the concatenation operation, and $\text{Conv}_{1\times1}(\cdot)$ denotes the 1×1 convolution kernel.

III. Experiments

A. Evaluation of Metrics

In the experiment, we adopt five evaluation metrics to assess the performance of the model: peak signal-to-noise ratio (PSNR) [40], spectral angle mapper (SAM) [41], normalized global error Erreur Relative Globale Adimensionnelle de Synthèse (ERGAS) [42], root-mean-square error (RMSE) [43], and structural similarity index (SSIM) [44]. PSNR is widely used to assess the performance of signal reconstruction; larger values indicate better image reconstruction. SAM measures the similarity of spectral angles; the smaller the SAM value, the better the result. ERGAS provides a comprehensive measure of the reconstructed images. A lower ERGAS value indicates better image reconstruction quality. RMSE computes the square difference between the GT and the reconstructed image. The smaller the RMSE value, the better the reconstruction image. SSIM evaluates the structural similarity between the fused image and the reference image. The higher the value of SSIM, the better the recovery result.

B. Datasets

1) CAVE Dataset: The CAVE dataset [45] was collected by Columbia University by using an Apogee Alta U260 camera. It contains 32 HSIs of $512 \times 512 \times 31$, with a wavelength range from 400 to 700 nm. The 21 images are chosen for training, and the other 11 images are used as the test set.

2) Harvard Dataset: Harvard dataset [46] has 50 HSIs. Each image size is $1392 \times 1040 \times 31$, and the wavelength range is 420–720 nm. The 40 images are selected for training, while the remaining images are used for testing.

3) Pavia Center Dataset: The Pavia Center dataset [47] was captured over Pavia. The size of the images is $1096 \times 1096 \times 102$, where the noise bands and the bands absorbed by water vapor are removed from the original 115 bands. In the experiment, the 512×512 area in the lower left corner is cropped for testing, and the remaining part is divided into 64×64 patches for training.

4) Chikusei Dataset: The Chikusei dataset [48] was collected from the agricultural and urban regions in Chikusei City, Japan. It consists of 128 spectral channels, the size is 2517×2335, and the wavelength range is 363–1018 nm. Six nonoverlapping patches of size 512×512 are selected as the test set, and the rest areas are divided into overlapping patches of size 64×64 for the training set.

In the experiment, the LR-HSIs are generated from the HR-HSIs by using a Gaussian blur with a standard deviation of 0.5, followed by a downsampling operation with a factor of 4. Meanwhile, the HR-MSIs are captured by using the camera's spectral response function.

C. Implementation Details

To evaluate the performance of our model, we compare our method with several state-of-the-art image fusion methods, including CSSNet [31], HSRNet [49], MCTNet [50], MDC-fusformer [32], AMSF [51], and Fusion-Mamba [36]. The experiment is trained by using the PyTorch framework with an NVIDIA GeForce RTX 4090 GPU. Adam optimizer is selected to train the model, and the initial learning rate is set to 0.0001.

D. Ablation Studies

In the experiment, we conduct some ablation experiments to show the effectiveness of the proposed method.

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TABLE I

ABLATION EXPERIMENTAL RESULTS OF OUR METHOD WITH OR WITHOUT SPAMB AND SPEMB MODULES

Module		CAVE					Harvard					Pavia Centre					Chikusei				
SpaMB	SpeMB	PSNR	SAM	ERGAS	RMSE	SSIM	PSNR	SAM	ERGAS	RMSE	SSIM	PSNR	SAM	ERGAS	RMSE	SSIM	PSNR	SAM	ERGAS	RMSE	SSIM
✗	✓	47.48	2.7308	1.3610	0.0041	0.9938	46.78	2.3336	2.1657	0.0049	0.9737	46.17	1.6088	0.9088	0.0051	0.9966	34.12	2.2096	3.7467	0.0111	0.9068
✓	✗	47.41	2.7154	1.3618	0.0042	0.9939	46.69	2.3600	2.1768	0.0050	0.9735	46.18	1.6092	0.9087	0.0051	0.9967	34.17	2.1671	3.6674	0.0110	0.9071
✗	✗	47.37	2.7341	1.3820	0.0042	0.9937	46.67	2.3775	2.1952	0.0050	0.9733	46.15	1.6068	0.9105	0.0051	0.9966	34.14	2.2138	3.7537	0.0111	0.9064
✓	✓	**47.68**	**2.6944**	**1.3451**	**0.0041**	**0.9940**	**46.96**	**2.3036**	**2.1340**	**0.0048**	**0.9741**	**46.20**	**1.5865**	**0.9020**	**0.0050**	**0.9967**	**34.50**	**2.0450**	**3.5382**	**0.0106**	**0.9122**

TABLE II

ABLATION EXPERIMENTAL RESULTS OF OUR METHOD WITH OR WITHOUT SPACROSS AND SPECROSS MODULES

Module		CAVE					Harvard					Pavia Centre					Chikusei				
SpaCross	SpeCross	PSNR	SAM	ERGAS	RMSE	SSIM	PSNR	SAM	ERGAS	RMSE	SSIM	PSNR	SAM	ERGAS	RMSE	SSIM	PSNR	SAM	ERGAS	RMSE	SSIM
✗	✓	47.32	2.7308	1.4029	0.0042	0.9938	46.34	2.4440	2.2638	0.0051	0.9720	44.96	1.6739	1.0713	0.0058	0.9963	34.25	2.1751	3.7039	0.0109	0.9068
✓	✗	47.27	2.7577	1.4320	0.0043	0.9938	46.43	2.4165	2.2595	0.0051	0.9720	44.81	1.6625	1.1176	0.0059	0.9963	34.25	2.1218	3.6388	0.0109	0.9081
✗	✗	46.95	2.8221	1.4716	0.0044	0.9935	46.33	2.4553	2.2942	0.0051	0.9724	44.83	1.6819	1.1127	0.0059	0.9963	34.15	2.1695	3.6777	0.0110	0.9068
✓	✓	**47.68**	**2.6944**	**1.3451**	**0.0041**	**0.9940**	**46.96**	**2.3036**	**2.1340**	**0.0048**	**0.9741**	**46.20**	**1.5865**	**0.9020**	**0.0050**	**0.9967**	**34.50**	**2.0450**	**3.5382**	**0.0106**	**0.9122**

TABLE III

ABLATION EXPERIMENTAL RESULTS OF THE SSLB MODULE

Module		CAVE					Harvard					Pavia Centre					Chikusei				
SpeMB	Laplace	PSNR	SAM	ERGAS	RMSE	SSIM	PSNR	SAM	ERGAS	RMSE	SSIM	PSNR	SAM	ERGAS	RMSE	SSIM	PSNR	SAM	ERGAS	RMSE	SSIM
✗	✓	47.25	2.7423	1.4249	0.0043	0.9937	46.41	2.4465	2.2601	0.0051	0.9720	45.35	1.6330	1.0233	0.0055	0.9965	34.26	2.1593	3.6301	0.0109	0.9083
✓	✗	47.16	2.7641	1.4101	0.0043	0.9937	46.31	2.4268	2.3051	0.0052	0.9724	44.98	1.7339	1.0510	0.0057	0.9960	34.15	2.1578	3.6440	0.0110	0.9059
✗	✗	46.97	2.7711	1.4402	0.0044	0.9936	46.20	2.4768	2.3273	0.0052	0.9717	44.93	1.7238	1.0727	0.0058	0.9961	34.10	2.1681	3.6461	0.0109	0.9046
✓	✓	**47.68**	**2.6944**	**1.3451**	**0.0041**	**0.9940**	**46.96**	**2.3036**	**2.1340**	**0.0048**	**0.9741**	**46.20**	**1.5865**	**0.9020**	**0.0050**	**0.9967**	**34.50**	**2.0450**	**3.5382**	**0.0106**	**0.9122**

1) Spatial Mamba (SpaMB) and Spectral Mamba (SpeMB): In the proposed network, SpaMB and SpeMB play a crucial role in the spatial–spectral feature extraction process. By scanning different spatial and spectral directions, SSM can effectively extract long-range spatial and spectral features. Table I shows the ablation results of the proposed method with or without the spatial mamba and spectral mamba on all datasets. Specifically, ✗-SpaMB denotes the proposed SSCM without spatial mamba. ✗-SpeMB denotes the proposed SSCM without spectral mamba. As shown in Table I, it is noted that the proposed SSCM with the SpaMB and the SpeMB can achieve better performance in comparison with other variants.

2) Spatial–Spectral Cross Mamba Block: To validate the effectiveness of the SSCM strategy, we conducted three experiments on all datasets. In Table II, ✗-SpaCross denotes the proposed SSCM with single spatial mamba instead of the spatial cross-fusion mamba between the LR-HSI features and the HR-MSI features. ✗-SpeCross denotes the proposed SSCM with a single spectral mamba instead of the spectral cross-fusion mamba between the LR-HSI features and the HR-MSI features. As illustrated in Table II, our SSCM with the spatial cross-fusion mamba and the spectral cross-fusion mamba can achieve better results than its variant methods. In comparison with the single mamba, our method improves the PSNR by 1.6% on average and decreases SAM, ERGAS, and RMSE by 5.7%, 9.1%, and 7.3% on average on all datasets. These results demonstrate that the SSCMB can effectively promote information interaction between the LR-HSI features and the HR-MSI features and improve the feature representation ability of the model.

3) Spatial–Spectral Learning Block: In the experiment, we evaluate the impact of the spatial–spectral learning module. In Table III, ✗-SpeMB denotes our SSLB without the spectral mamba branch, ✗-Laplace represents our SSLB without the Laplace operator branch. As shown in Table III, the experimental results show that our method with a three-branch structure can obtain comparable experimental performance. In comparison with the variant method with a SpaMB, our method improves the PSNR by 1.7% on average and decreases SAM, ERGAS, and RMSE by 6.1%, 7.9%, and 7.2% on average on all datasets. These results indicate that SSLB can effectively improve the performance of the reconstructed images.

To further demonstrate the effectiveness of the SSLB module, in Fig. 5, we visualize the feature maps from the three branches of SSLB and the final output, i.e., $F_{\text{Lap}}(x)$, $F_{\text{Spa}}(x)$, $F_{\text{Spe}}(x)$, and $\text{SSLB}(x)$. Each feature map is one of the feature channels in the output features. From Fig. 5, in comparison with other variant methods, the features derived from the SSLB

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TABLE IV

ABLATION EXPERIMENTAL RESULTS OF THE CFMB MODULE WITH OR WITHOUT HR-MSI FEATURES

Module		CAVE					Harvard					Pavia Centre					Chikusei				
$F_{HR\text{-}MSI_1}$	$F_{HR\text{-}MSI_2}$	PSNR	SAM	ERGAS	RMSE	SSIM	PSNR	SAM	ERGAS	RMSE	SSIM	PSNR	SAM	ERGAS	RMSE	SSIM	PSNR	SAM	ERGAS	RMSE	SSIM
✗	✓	46.29	2.7609	1.5396	0.0045	0.9933	46.84	2.3429	2.1632	0.0049	0.9735	45.71	1.6714	0.9549	0.0052	0.9964	34.17	2.2210	3.6843	0.0110	0.9066
✓	✗	47.52	2.7400	1.3713	0.0041	0.9937	45.92	2.3417	2.3573	0.0054	0.9726	40.33	2.4212	1.7449	0.0097	0.9890	33.89	2.1511	3.9041	0.0114	0.9006
✗	✗	46.11	2.8865	1.6667	0.0049	0.9927	44.96	2.7034	2.4950	0.0060	0.9678	38.31	2.8296	2.2021	0.0122	0.9832	33.71	2.2761	3.7505	0.0114	0.9019
✓	✓	**47.68**	**2.6944**	**1.3451**	**0.0041**	**0.9940**	**46.96**	**2.3036**	**2.1340**	**0.0048**	**0.9741**	**46.20**	**1.5865**	**0.9020**	**0.0050**	**0.9967**	**34.50**	**2.0450**	**3.5382**	**0.0106**	**0.9122**

TABLE V

RECONSTRUCTION PERFORMANCE OF VARIOUS METHODS ON THE CAVE, HARVARD, PAVIA CENTER, AND CHIKUSEI DATASETS

Method	CAVE					Harvard					Pavia Centre					Chikusei				
	PSNR	SAM	ERGAS	RMSE	SSIM	PSNR	SAM	ERGAS	RMSE	SSIM	PSNR	SAM	ERGAS	RMSE	SSIM	PSNR	SAM	ERGAS	RMSE	SSIM
CSSNET [31]	36.54	4.6543	4.7297	0.0148	0.9675	38.37	3.1291	4.2976	0.0129	0.9385	38.97	2.5882	1.9384	0.0113	0.9856	32.78	2.6489	4.3889	0.0130	0.8831
HSRNET [49]	46.23	3.6473	1.6117	0.0048	0.9917	46.39	2.4808	2.2015	0.0051	0.9740	40.30	2.1591	1.3549	0.0097	0.9912	33.57	2.2256	3.6728	0.0118	0.8947
MDC-NET [32]	44.36	3.6518	2.0689	0.0063	0.9893	44.36	2.8012	2.4303	0.0067	0.9723	45.09	1.7743	0.9884	0.0056	0.9963	33.84	2.2942	3.7242	0.0114	0.9004
MCT-NET [50]	45.56	3.241	1.8663	0.0055	0.9902	45.46	2.6381	2.3562	0.0059	0.9726	43.38	2.0695	1.2196	0.0068	0.9951	**34.55**	2.2659	3.6518	**0.0105**	0.9100
Fusion Mamba [36]	46.87	**2.6745**	1.5079	0.0046	0.9932	46.33	2.3352	**2.1048**	0.0052	**0.9744**	45.41	1.6875	1.0020	0.0054	0.9964	33.54	2.1183	3.6056	0.0119	0.9001
AMSF [51]	41.01	4.2629	3.0108	0.0094	0.9865	41.65	3.6530	3.1164	0.0091	0.9666	44.11	1.9213	1.1598	0.0063	0.9956	34.22	2.4773	3.8257	0.0110	0.9048
SSCM	**47.68**	2.6944	**1.3451**	**0.0041**	**0.9940**	**46.96**	**2.3036**	2.1340	**0.0048**	0.9741	**46.20**	**1.5865**	**0.9020**	**0.0050**	**0.9967**	34.50	**2.0450**	**3.5382**	0.0106	**0.9122**

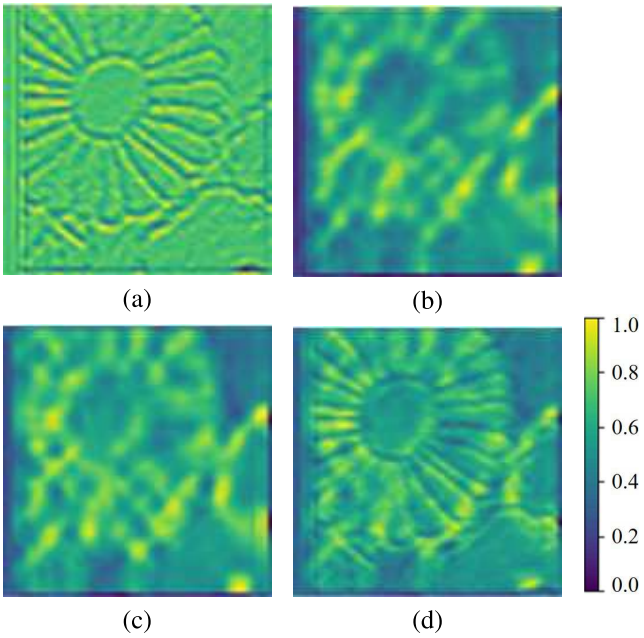


Fig. 5. Visual feature maps from the three branches of SSLB and the final output. (a) $F_{\text{Lap}}(x)$. (b) $F_{\text{Spa}}(x)$. (c) $F_{\text{Spe}}(x)$. (d) SSLB(x).

with the Laplacian operator and spectral mamba have richer image structure details information and show stronger feature learning ability.

4) Cross-Fusion Mamba Block: In the experiment, we analyze the effectiveness of the CFMB. In Table IV, ✗-$F_{\text{HR}-\text{MSI}_1}$ denotes the proposed CFMB without the enhanced high-resolution multispectral features $F_{\text{HR}-\text{MSI}_1}$ features, and ✗-F_{HR-MSI_2} denotes the proposed CFMB without the enhanced high-resolution multispectral features $F_{\text{HR}-\text{MSI}_2}$. As shown in Table IV, in compassion with the other modules, our method can obtain better experimental results and show good feature learning ability.

E. Experimental Results

1) CAVE Dataset: The experimental results are shown in Table V. It can be seen that our SSCM can achieve better performance than other state-of-the-art HSI fusion methods. Specifically, compared to the second-best method, our method improves PSNR by 1.7% and decreases ERGAS and RMSE by 10.7% and 10.9%, respectively. Furthermore, Fig. 6 illustrates the evaluation results for each band across different fusion methods on the CAVE test set. The experimental results indicates that our method achieves comparable results in each spectral band compared to other image fusion methods. In addition, in Fig. 7, we present the pseudo-color images (R-24, G-16, and B-8) of different image fusion methods. From the visual results in Fig. 7, we can find that our SSCM can obtain better image reconstruction performance in comparison with other image fusion methods. In terms of difference images, our method also achieves small difference results than other methods. The above results show that our method has good learning ability in spatial and spectral reconstructions.

2) Pavia Center Dataset: The experimental results are presented in Table V. From Table V, it is noted that the proposed method yields superior results in comparison with other fusion methods in all metrics. Specifically, compared to MCT-NET, our PSNR is improved by 6.5%, and when compared to Fusion Mamba, our PSNR is also improved by 1.7%. This shows that our method has the ability to preserve spatial detail information. In comparison with the second-best method, our method decreases by 6.0% and 8.7% in terms of SAM and ERGAS, respectively. In addition, Fig. 8 presents the reconstructed pseudo-color images (R-67, G-29, and B-1) of different methods. From the visual results of Fig. 8, it can

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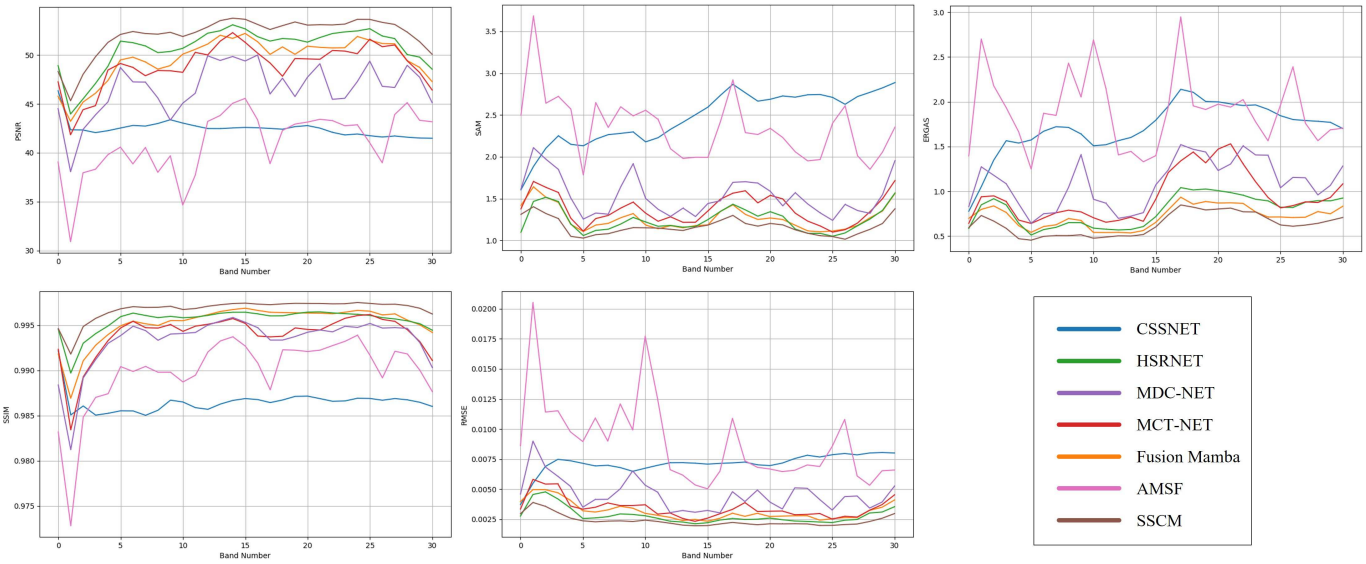


Fig. 6. Experimental results of our SSCM and other methods on the CAVE test set.

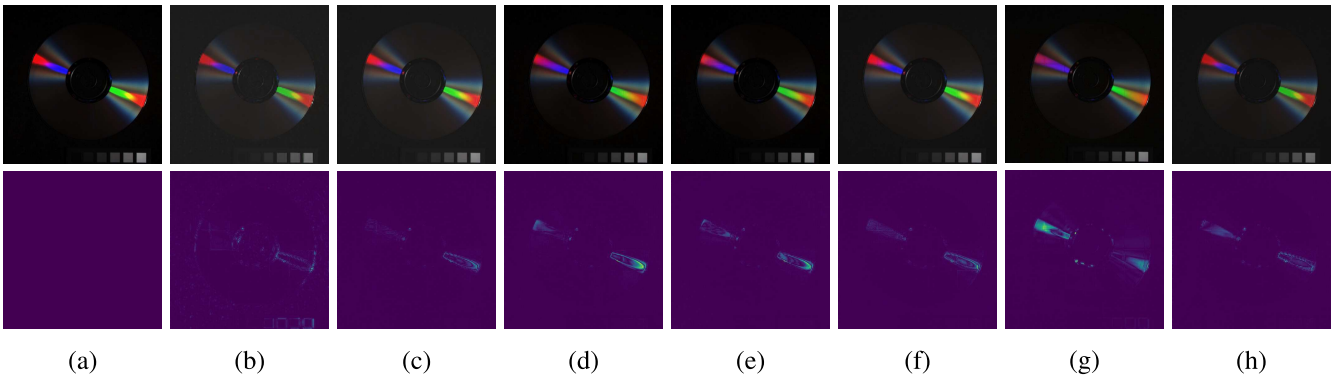


Fig. 7. Reconstruction results of various methods on the CAVE dataset. The first row shows the pseudo-color images (R-24, G-16, and B-8), while the second row shows the difference images in relation to the pseudo-color image and GT. (a) GT. (b) CSSNET. (c) HSRNET. (d) MDC-NET. (e) MCT-NET. (f) Fusion Mamba. (g) AMSF. (h) SSCM.

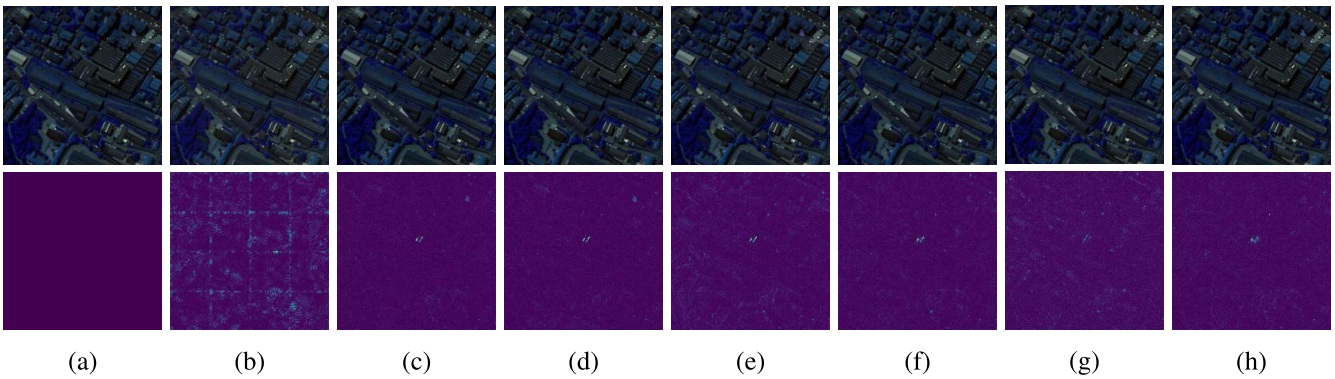


Fig. 8. Reconstruction results of various methods on the Pavia Center dataset. The first row shows the pseudo-color images (R-67, G-29, and B-1), while the second row shows the difference images in relation to the pseudo-color image and GT. (a) GT. (b) CSSNET. (c) HSRNET. (d) MDC-NET. (e) MCT-NET. (f) Fusion Mamba. (g) AMSF. (h) SSCM.

be seen that our SSCM can generate better visual results than other image fusion methods.

3) Harvard and Chikusei Datasets: The experimental results are presented in Table V. It can be found that our method achieves comparable experimental performance. Specifically, on the Harvard dataset, in comparison with Fusion Mamba, our method improves PSNR by 1.3% and decreases SAM by 1.3%. In addition, Figs. 9 and 10 show the

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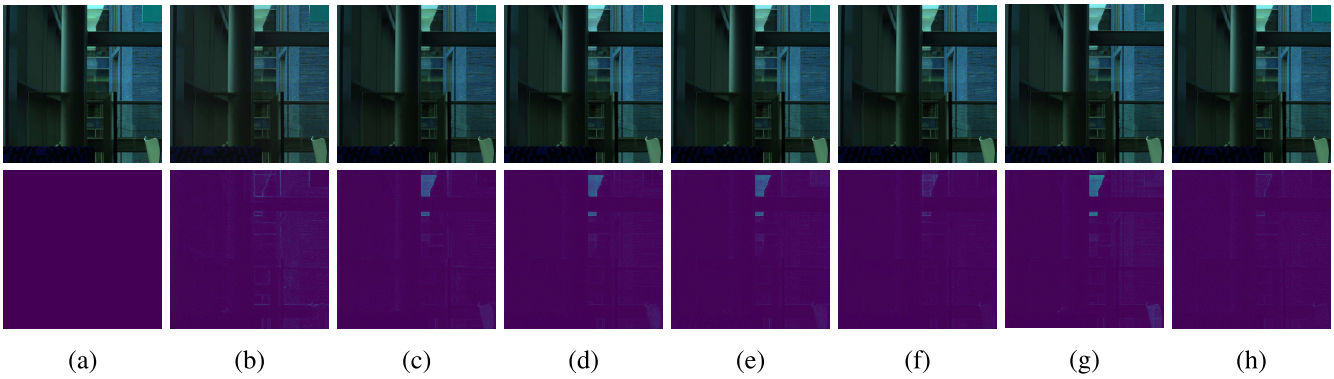


Fig. 9. Reconstruction results of various methods on the Harvard dataset. The first row shows the pseudo-color images (R-24, G-16, and B-8), while the second row shows the difference images in relation to the pseudo-color image and GT. (a) GT. (b) CSSNET. (c) HSRNET. (d) MDC-NET. (e) MCT-NET. (f) Fusion Mamba. (g) AMSF. (h) SSCM.

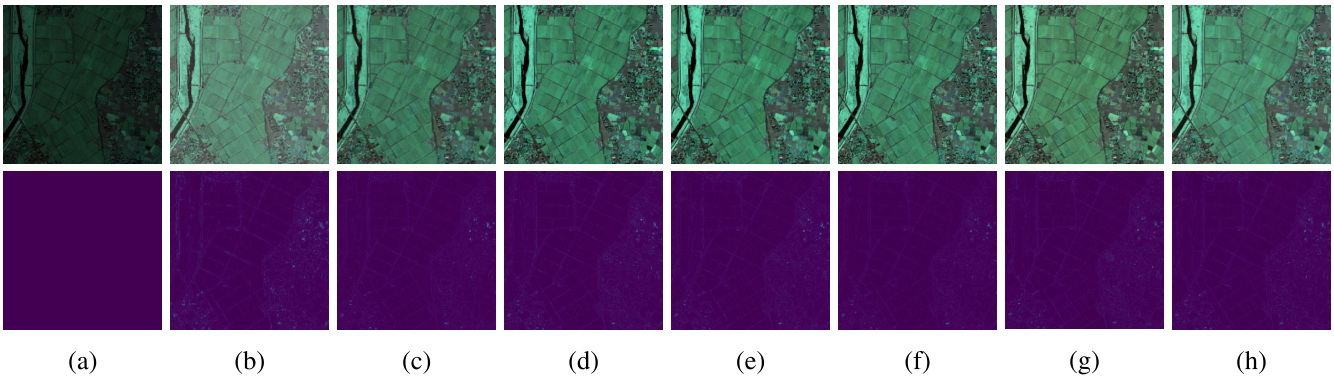


Fig. 10. Reconstruction results of various methods on the Chikusei dataset. The first row shows the pseudo-color images (R-119, G-90, and B-69), while the second row shows the difference images in relation to the pseudo-color image and GT. (a) GT. (b) CSSNET. (c) HSRNET. (d) MDC-NET. (e) MCT-NET. (f) Fusion Mamba. (g) AMSF. (h) SSCM.

TABLE VI
RECONSTRUCTION PERFORMANCE OF DIFFERENT METHODS TRAINED ON THE CAVE DATASET AND TESTED ON THE HARVARD DATASET

Method	Harvard				
	PSNR	SAM	ERGAS	RMSE	SSIM
CSSNET [31]	39.21	3.2232	4.1300	0.0116	0.9446
HSRNET [49]	44.28	3.0711	2.8537	0.0064	0.9656
MDC-NET [32]	39.40	5.0465	11.2406	0.0118	0.9465
MCT-NET [50]	37.83	5.9261	17.2241	0.0141	0.9307
Fusion Mamba [36]	44.32	2.9422	4.1928	0.0064	0.9566
AMSF [51]	34.88	8.6254	17.8838	0.0200	0.9228
SSCM	**45.22**	**2.8233**	**2.7528**	**0.0057**	**0.9667**

TABLE VII
RECONSTRUCTION PERFORMANCE OF DIFFERENT METHODS TRAINED ON THE HARVARD DATASET AND TESTED ON THE CAVE DATASET

Method	CAVE				
	PSNR	SAM	ERGAS	RMSE	SSIM
CSSNET [31]	34.47	5.3893	6.2124	0.0188	0.9487
HSRNET [49]	33.40	19.0296	6.2258	0.0212	0.9172
MDC-NET [32]	31.86	21.3696	7.6994	0.0252	0.9088
MCT-NET [50]	32.27	19.7418	7.5211	0.0243	0.9057
Fusion Mamba [36]	37.88	8.0911	3.8512	0.0124	0.9647
AMSF [51]	28.89	26.0081	10.6767	0.0355	0.8876
SSCM	**40.16**	**5.0977**	**3.1699**	**0.0095**	**0.9737**

reconstructed pseudo-color images of different methods. As shown in Figs. 9 and 10, it can be seen that our method can achieve better image fusion results than other methods.

F. Generalizability Experiment

To demonstrate the generalization ability of different methods, we conduct generalization experiments on the CAVE and Harvard datasets, respectively. Table VI shows the experimental results of different methods trained on the CAVE dataset and then tested on the Harvard dataset, and Table VII shows the experimental results of different methods trained on the Harvard dataset and then tested on the CAVE dataset. As presented in Tables VI and VII, our method achieves better performance than other methods on all evaluation metrics. In addition, we show visual results of different methods with the pseudo-color images (R-24, G-16, and B-8). In Fig. 11, it can be seen that our method can produce smaller error results compared with other fusion methods, which demonstrates that our method has better generalization ability.

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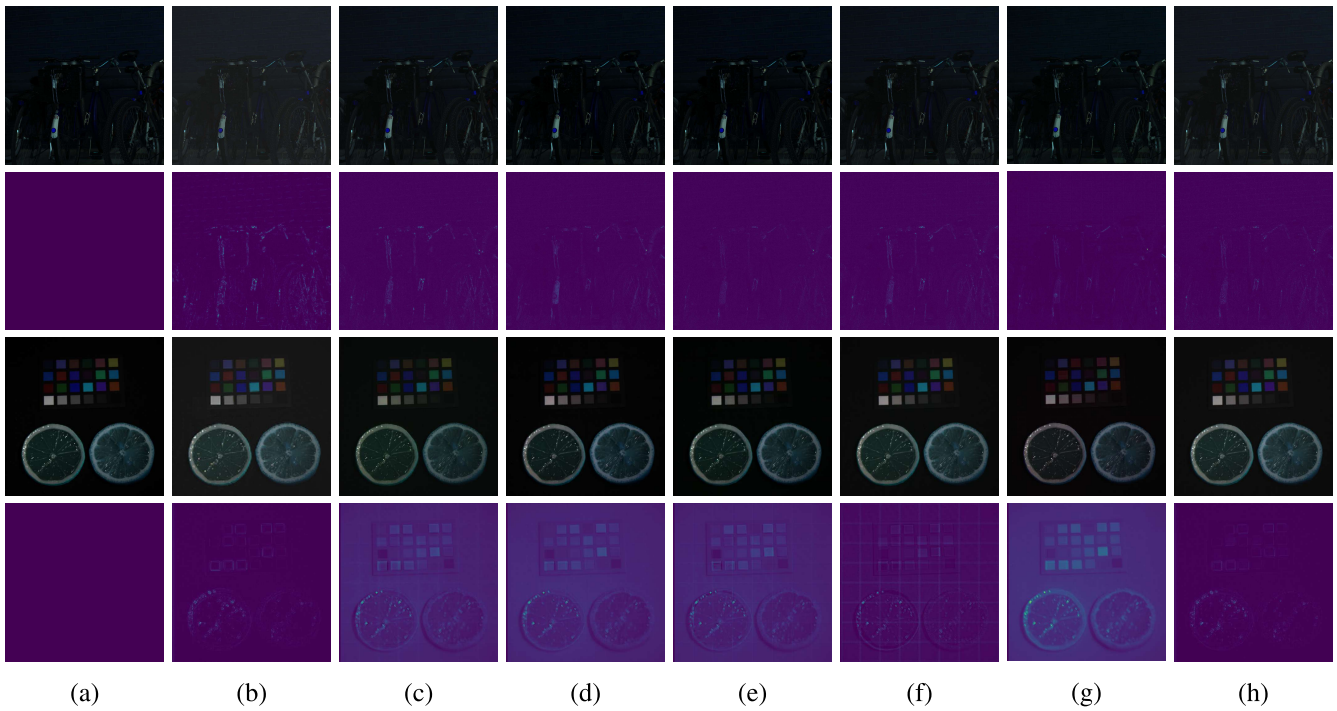


Fig. 11. Reconstruction results of various methods on the Harvard and CAVE datasets. The first row shows the pseudo-color images (R-24, G-16, and B-8) on the Harvard dataset, while the second row shows the difference images in relation to the pseudo-color image and GT on the Harvard dataset. The third row shows the pseudo-color images (R-24, G-16, and B-8) on the CAVE dataset, while the fourth row shows the difference images in relation to the pseudo-color image and GT on the CAVE dataset. (a) GT. (b) CSSNET. (c) HSRNET. (d) MDC-NET. (e) MCT-NET. (f) Fusion Mamba. (g) AMSF. (h) SSCM.

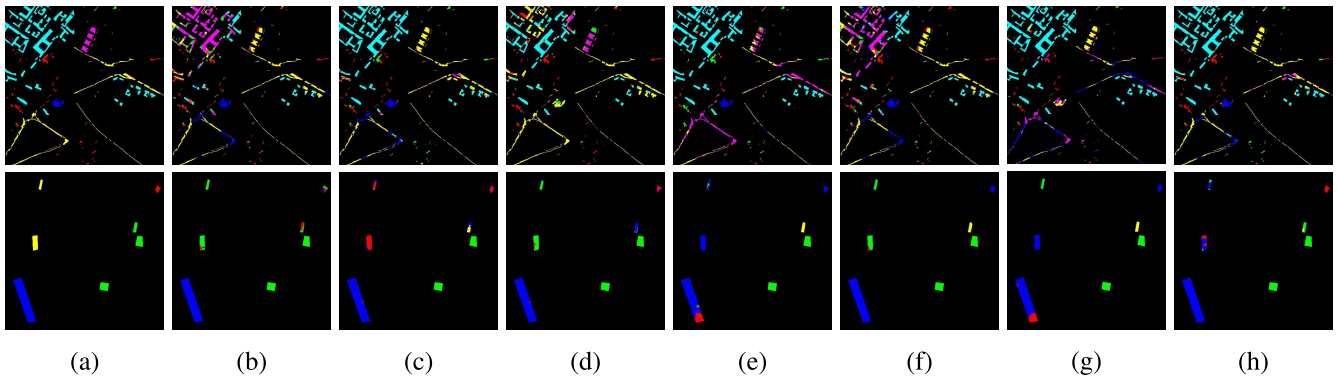


Fig. 12. Classification map obtained from the reconstructed images using different methods on the HybridSN model. The first row shows the classification maps on the Pavia Center dataset, and the second row shows the classification maps on the Chikusei dataset. (a) GT. (b) CSSNET. (c) HSRNET. (d) MDC-NET. (e) MCT-NET. (f) Fusion Mamba. (g) AMSF. (h) SSCM.

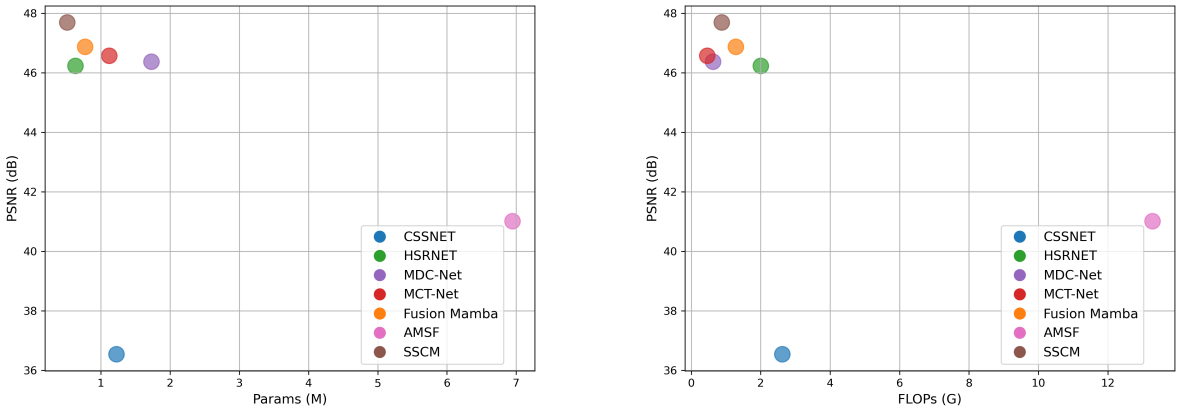
G. Classification Experiments

To further explore the potential of our method, we conduct HSI classification experiments of different methods on the Pavia and Chikusei datasets by using HybridSN [52]. Tables VIII and IX show the results of overall accuracy (OA), average accuracy (AA), and Kappa coefficient (Kappa) of different methods on these two datasets, respectively. In terms of OA, AA, and Kappa, our method improves by 4.7%, 3.0%, and 8.6% on the Pavia dataset, and by 4.7%, 12.5%, and 7.8% on the Chikusei dataset in comparison with the second-best method, respectively. In addition, the classification visualization results are shown in Fig. 12. It can be found that our SSCM can achieve comparable experimental performance, which further demonstrates that our method has better spatial and spectral learning abilities.

H. Analysis of Computational Complexity

To further show the effectiveness of our method, in Fig. 13, we report two metrics on different model's computational complexity, i.e., the number of model parameters and the number of floating-point operations (FLOPs). We use the image from CAVE with an upscaling factor $\times 4$ to compute FLOPs, where the size of HR-HSI images is $64 \times 64 \times 31$ pixels. As shown in Fig. 13, in comparison with other state-of-the-art methods, our method can achieve comparable results in terms of model performance and computational complexity.

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(a) (b)

Fig. 13. Comparisons of the PSNR, FLOPs, and the number of parameters. (a) PSNR versus parameters and (b) PSNR versus FLOPs.

TABLE VIII
CLASSIFICATION ACCURACY ON HYBRIDSN MODEL USING DIFFERENT METHODS ON THE PAVIA CENTER DATASET

Method	Pavia Centre		
	OA	AA	Kappa
CSSNET [31]	0.3827	0.4604	0.2530
HSRNET [49]	0.8072	0.6754	0.6695
MDC-NET [32]	0.7146	0.5593	0.5674
MCT-NET [50]	0.7864	0.6640	0.6348
Fusion Mamba [36]	0.4125	0.4563	0.2678
AMSF [51]	0.7410	0.4681	0.5519
SSCM	**0.8468**	**0.6963**	**0.7327**

TABLE IX
CLASSIFICATION ACCURACY ON HYBRIDSN MODEL USING DIFFERENT METHODS ON THE CHIKUSEI DATASET

Method	Chikusei		
	OA	AA	Kappa
CSSNET [31]	0.8025	0.5654	0.6529
HSRNET [49]	0.7971	0.6481	0.6544
MDC-NET [32]	0.8017	0.6464	0.6429
MCT-NET [50]	0.6633	0.4059	0.4068
Fusion Mamba [36]	0.7775	0.4580	0.6011
AMSF [51]	0.6742	0.4106	0.4304
SSCM	**0.8421**	**0.7403**	**0.7100**

It further demonstrates that our SSCM has a good tradeoff between model complexity and performance.

IV. CONCLUSION

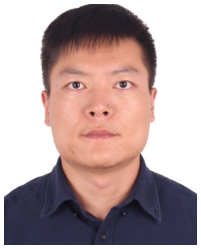
In this article, we propose an SSCM network for the LR-HSI and HR-MSI fusion. Specifically, an SSCMB is introduced to effectively extract and complement informative spatial and spectral feature information between the LR-HSI and HR-MSI. In addition, we present a U-shaped structure SSLB and CFMB, enabling the network to extract important long-range dependencies and high-frequency texture and edge features at multiple scales. Various experimental results demonstrate that our SSCM obtains better image restoration results and comparable generalization in comparison with the other state-of-the-art image fusion methods.

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Huapeng Wu received the Ph.D. degree from Nanjing University of Science and Technology, Nanjing, China, in 2021.

He is currently a Lecturer with the School of Computer Science, Nanjing Audit University, Nanjing. His research interests include hyperspectral image processing, super-resolution, and deep learning.



Zhiyu Sun received the B.S. degree in software engineering from Xuzhou University of Technology, Xuzhou, China, in 2023. He is currently pursuing the M.S. degree in computer science and technology with Nanjing Audit University, Nanjing, China.

His research interests include hyperspectral image processing.

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Jiaqiang Qi received the B.S. degree in software engineering from Nanjing Audit University, Nanjing, China, in 2023, where he is currently pursuing the M.S. degree in computer science and technology.

His research interests include hyperspectral image processing.



Yang Xu (Member, IEEE) received the B.Sc. degree in applied mathematics and the Ph.D. degree in pattern recognition and intelligence systems from Nanjing University of Science and Technology (NUST), Nanjing, China, in 2011 and 2016, respectively.

He is currently an Associate Professor with the School of Computer Science and Engineering, NUST. His research interests include hyperspectral image classification, hyperspectral detection, image processing, and machine learning.



Tianming Zhan (Member, IEEE) received the B.S. and M.S. degrees in applied mathematics from the School of Math and Statistics, Nanjing University of Information Science and Technology, Nanjing, China, in 2006 and 2009, respectively, and the Ph.D. degree in pattern recognition and intelligence systems from the School of Computer Science and Engineering, Nanjing University of Science and Technology, Nanjing, in 2013.

He is currently a Professor with the School of Computer and Jiangsu Modern Intelligent Audit Integrated Application Technology Engineering Research Center, Nanjing Audit University, Nanjing. His research interests include hyperspectral image processing, machine learning, and data analysis.



Zhihui Wei (Member, IEEE) received the B.Sc., M.Sc., and Ph.D. degrees from Southeast University, Nanjing, China, in 1983, 1986, and 2003, respectively.

He is currently a Professor and a Doctoral Supervisor with Nanjing University of Science and Technology, Nanjing. His research interests include mathematical image modeling, multiscale analysis, video and image coding and compressing, watermarking and steganography, speech and audio processing, mathematical methods, and machine learning techniques in data science, such as multiscale geometrical analysis, manifold learning and regularization, optimization method for big data processing, and deep learning.

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